perhaps as far again from Neptune as Neptune is from the Sun. The Dutch-born American astronomer Gerard Kuiper advanced the idea that in the cold depths of space, beyond the orbit of Neptune, there reside frozen leftovers from the formation of the solar system. Without a massive planet upon which to fall, most of these comets will orbit the Sun for billions more years. As is true for the asteroid belt, some objects of the Kuiper belt travel on eccentric paths that cross the orbits of other planets. Pluto and its ensemble of siblings called Plutinos cross Neptune's path around the Sun. Other Kuiper belt objects plunge all the way down to the inner solar system, crossing planetary orbits with abandon. This subset includes Halley, the most famous comet of them all.

Far beyond the Kuiper belt, extending halfway to the nearest stars, lives a spherical reservoir of comets called the Oort cloud, named for Jan Oort, the Dutch astrophysicist who first deduced its existence. This zone is responsible for the long-period comets, those with orbital periods far longer than a human lifetime. Unlike Kuiper belt comets, Oort cloud comets can rain down on the inner solar system from any angle and from any direction. The two brightest of the 1990s, comets Hale-Bopp and Hyakutake, were both from the Oort cloud and are not coming back anytime soon.

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If we had eyes that could see magnetic fields, Jupiter would look ten times larger than the full Moon in the sky. Spacecraft that visit Jupiter must be designed to remain unaffected by this powerful force. As the English physicist Michael Faraday demonstrated in the 1800s, if you pass a wire across a magnetic field you generate a voltage difference along the wire's length. For this reason, fast-moving metal space probes will have electrical currents induced within them. Meanwhile, these currents generate magnetic fields of their own that interact with the ambient magnetic field in ways that retard the space probe's motion.

Last I had kept count, there were fifty-six moons among the planets in the solar system. Then I woke up one morning to learn that another dozen had been discovered around Saturn. After that incident, I decided to no longer keep track. All I care about now is whether any of them would be fun places to visit or to study. By some measures, the solar system's moons are much more fascinating than the planets they orbit.